BIOCHEMISTRY CUMULATIVE EXAMINATION

December 1, 1990

A liver cell suspension is depleted of carbohydrates and amino acids down to low levels. Then, glutamic acid labeled with C-14 in carbon atom number 4 is added to the liver cell suspension. After a short period of time, aspartic acid begins to appear in the liver cell, and is labeled with C-14.

A) (60pts) Write a complete enzymatic pathway for the synthesis of aspartic acid in these mitochondria using glutamic acid as the source of carbon and nitrogen.

B) (40pts) Which carbon atoms of aspartic acid will first be labeled with C-14. Show your work and explain!

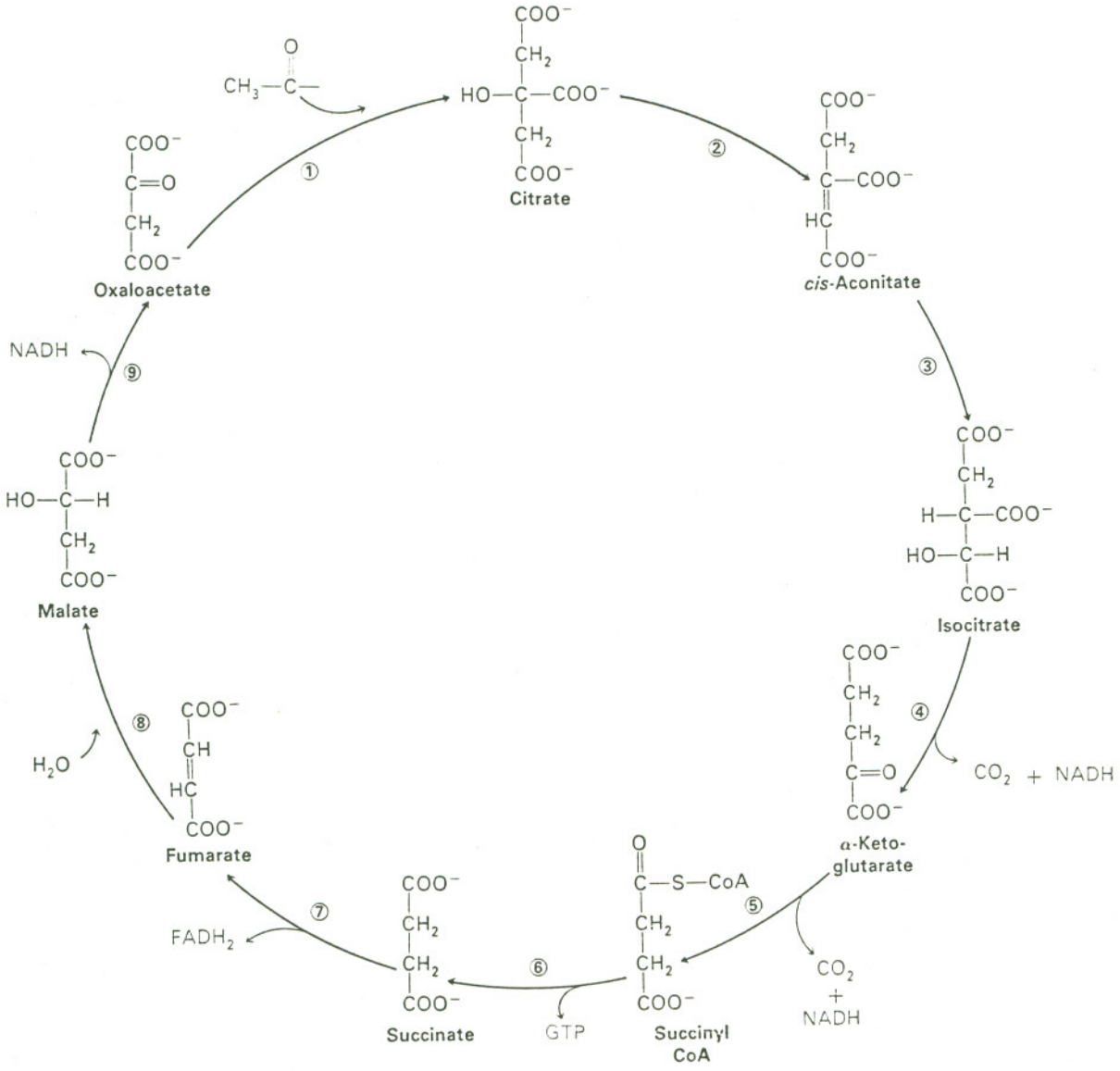

Figure 16-6
Citric acid cycle.

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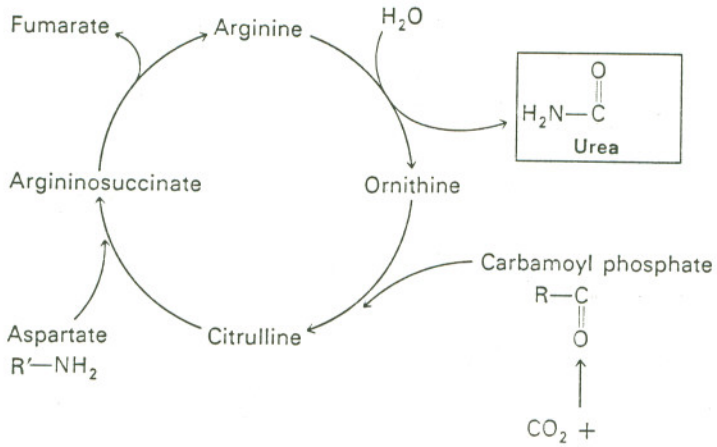
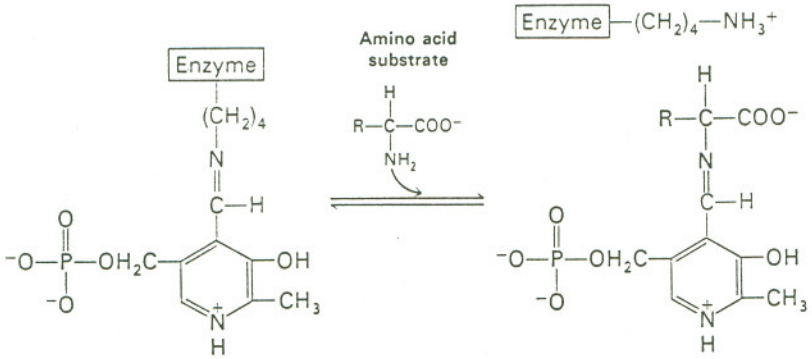
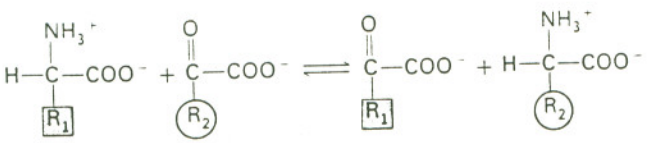
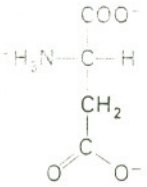


Figure 21-6
The urea cycle.

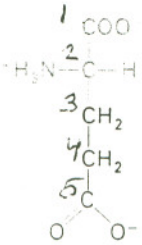


$$\alpha\text{-Amino acid} + \underset{(\text{or NADP}^+)}{\text{NAD}^+} + H_2O \rightleftharpoons \alpha\text{-keto acid} + NH_4^+ + \underset{(\text{or NADPH})}{\text{NADH}} + H^+$$





Aspartate
(Asp, D)



Glutamate
(Glu, E)